

The background is a dark blue gradient. It is decorated with several vertical white lines of varying lengths and numerous small squares in shades of pink, orange, and teal, scattered across the slide.

Xficient

Tackling Healthcare Administrative Waste with Generative AI

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Problem Statement

Healthcare Spending :

- Accounts for 15–20% of US GDP
- 2023: \$4.8T (approx. 17.5% of US GDP)

Administrative Spending :

- 20–30% of total spending is on administrative functions
- 50% of this is wasted
- Annual waste: \$480–720B (10–15% of healthcare spending)

Proprietary Data Formats :

- Standard Electronic Data Interchange (EDI) formats: 834, 835, 837
- Data exchange involves payers, providers, pharmacies, etc.

Current Issues :

- Differences in standards, protocols, and data models
- Mapping between systems is complex and time-consuming

Effort Required :

- Months of data analyst and software engineering time for each exchange

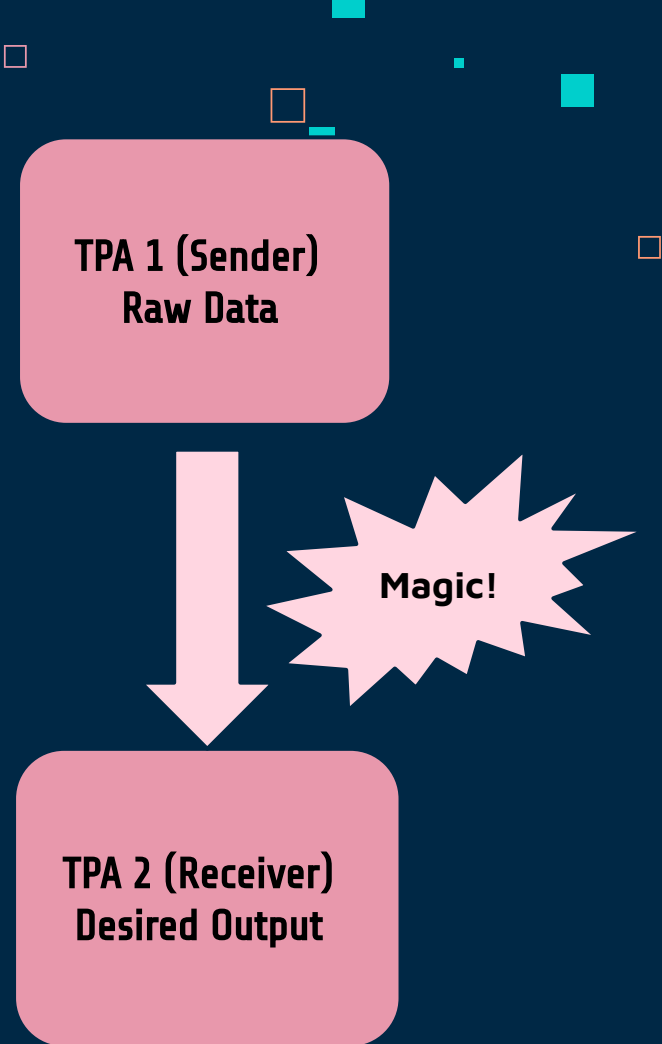
Mapping Problem

Scenario :

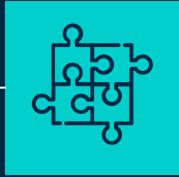
- *TPA 1 (Sender)* and *TPA 2 (Receiver)* need to exchange information
- Both TPAs have similar data but different database schemas

Objective :

- Build a semantic mapping capability to translate between TPA 1's and TPA 2's databases



Solution!



01

Data Extraction



02

Mapping
Transformation



03

Form Population

Input and Output

Input Data

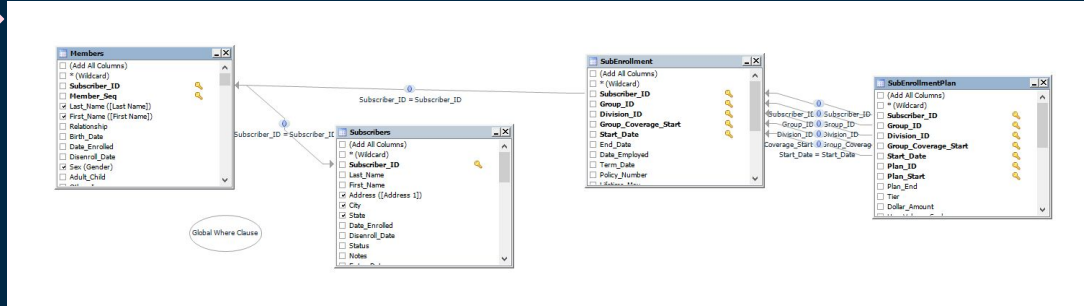
1. **TPA 1 Database Dump** :
 - Contains raw data from TPA 1's system
2. **Member Data Files** :
 - Includes CSV files like Relationship.csv, Tier.csv, etc.
3. **Helper Mapping Files** :
 - Provides mappings and organization information
4. **TPA 2 Companion Guide** :
 - "Layout.pdf" detailing how TPA 2 expects data to be formatted

Desired Output

1. **Filled-Out Form** :
 - Properly completed form (e.g., 834) based on TPA 1's data and TPA 2's Companion Guide

Current Solution

- Uses the specific relationship model
- Extracts data from the files
- Analyzes and stores important information using common data structures
- Follows TPA 2 companion guide to map data correctly to the new form
- Outputs the formatted data in the excel file



Pro's and Con's of Solution

Pros

Automated Mapping :

- Reduces manual effort in translating data between different formats

Scalability :

- Solution can be scaled to handle large datasets and multiple TPAs

Flexibility :

- Accommodates variations in how different TPAs use forms

Cons

Data Quality Dependency :

- The effectiveness of the mapping depends on the quality and completeness of input data

Complexity of Mapping :

- Mapping between different database schemas and form expectations can be complex

Missing Fields

- Doesn't see the relationship between the members and their "Dependent Coverage Code"

Future Improvements

Using Generative AI to Automate Mapping

- Train a Generative AI model to recognize patterns and structures within the Companion Guide and apply transformations that map data according to the receiver's specifications
- Automate the process of translating data from one database schema (e.g., TPA 1) into the standardized form (e.g., 834) and format it to meet the expectations of the receiving TPA (e.g., TPA 2)

Accurate Dependent Coverage Mapping

- AI will correctly assign the dependent coverage code for each member by analyzing the relationship between each member and the subscriber
- Ensures the coverage is accurately applied for dependents (e.g., spouse, children) based on their specific relationship and eligibility rules

Future Outlook and Commitment

Importance of the Solution

Reducing administrative waste can save **billions** of dollars annually

Commitment to Future Development

Continuous model improvements to handle **increased complexity** and **new data formats**

Collaborating with data scientists and software engineering to tackle this problem